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1. Please choose one What is the maximum depth of recursive calls a function may make?

- a) 1
- b) 2
- c) n (where n is the argument)

d) There is no fixed maximum.

2. The difference between a binary tree and a binary search tree is that a binary search tree has?

a) two children per node whereas a binary tree can have none, one, or two children per node.

- b) in binary search tree nodes are inserted based on the values they contain.
- c) in binary tree nodes are inserted based on the values they contain.
- d) none of these.

3. Which of the following is NOT include in the definition of Data Structure?

- a) Data Management
- b) Data Organization
- c) Data Storage Format

d) Data Security

4. A queue where the de-queue operation depends not on FIFO, is called a priority queue?

- a) False

b) True

5. _____ is special data type to store the same data type?

a) An Array

- b) A List
- c) A Program
- d) A Function

6. The data of the problem is of 2GB and the hard disk is of 1GB capacity, to solve this problem we should?

- a) Use better data structures
- b) Increase the hard disk space**
- c) Use the better algorithm
- d) Use as much data as we can store on the hard disk

7. In an array list the current element is ?

- a) The middle element
- b) The first element**
- c) The last element
- d) The element where the current pointer points to

8. Which of the following function used is to move the pointer at the last element of List Data Structure?

- a) start()
- b) tail ()**
- c) length ()
- d) get ()

9. Which of the following is not included in the data structure?

- a) Space for each item
- b) Time to perform
- c) Programming effort
- d) None of the above**

10. Consider the function X as under

```
int X (int& Value)
{
    return Value;
}
```

Now a and b are integers in a calling function. Which one of the following is a valid call to the above function X?

- a) a=X (b);
- b) a = X (&b);**
- c) a = X (*b);
- d) None of the given options

11. "+" is a operator. www.thundershare.net ?

- a) Unary
- b) Binary**
- c) Ternary
- d) None of the above

12. In the call by value methodology, a copy of the object is passed to the called function?

- a) False
- b) True**

13. Each node in doubly link list has?

- a) 4 Pointer
- b) 3 Pointer
- c) 2 Pointer**
- d) 1 Pointer

14. The tree data structure is a?

- a) Linear data structure

- b) Graphical data structure
- c) Non-linear data structure**
- d) Data structure like queue

15. Doubly Linked List always has one NULL pointer?

- a) True
- b) False**

16. _____ is a collection of Element arranged in a linear order?

- a) Queue
- b) List
- c) Stack**
- d) Array

17. Suppose n is the number of nodes in a complete Binary Tree then maximum steps required for a search operation are?

- a) $\log_2(n+1)-1$**
- b) $\log_2(n+1)$
- c) $\log_2(n)-1$
- d) $\log_2(n)$

19. A subscript of an array may be an integer or an integer expression?

- a) True**
- b) False

20. Which of the following operator evaluates first?

- a) Addition
- b) Multiplication

c) Division

d) Exponentiation

21. In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?

a) At the head

b) At the tail

c) After all other entries that are greater than the new entry.

d) After all other entries that are smaller than the new entry.

22. The expression $AB+C*$ is called?

a) Prefix expression.

b) Postfix expression.

c) Infix expression.

d) None of these

23. The operation for adding an entry to a stack is traditionally called?

a) add

b) append

c) insert

d) push

24. The following statements are related to queues?

a) The last item to be added to a queue

b) The first item to be removed from queue

c) The queue used FIFO Strategy

d) All of the above

25. We access elements in AVL Tree in?

- a) Linear way only
- b) Non Linear way only**
- c) Both linear and non linear ways
- d) None of the given option

26. Which one is a self- referential data type?

- a) Stack
- b) Queue
- c) Link list**
- d) All of these

27. "--" is a _____ operator.

- a) Unary
- b) Binary**
- c) Ternary
- d) None of the above

28. _____ is a binary tree where every node has a value, every node's left sub tree contains only values less than or equal to the node's value, and every node's right subtree contains only values that are greater then or equal?

- a) Strictly Binary Tree**
- b) Binary Search tree
- c) AVL tree
- d) All of these

29. Suppose we have a circular array implementation of the queue class, with ten items in the queue stored at data[2] through data[11]. The CAPACITY is 42, i.e., the array has been

declared to be of size 42. Where does the push member function place the new entry in the array?

- a) data[1]
- b) data[2]
- c) data[11]
- d) data[12]**

30. A Compound Data Structure is the data structure which can have multiple data items of same type or of different types. Which of the following can be considered compound data structure?

- a) Arrays
- b) LinkLists
- c) Binary Search Trees**
- d) All of the given options

31. AVL tree is developed by_____?

- a) Adelson
- b) Velaskii
- c) Landis
- d) Adelson-Velaskii and Landis**

32. How to access the first element of the array 1?

- a) array[0]
- b) arr1[0]1
- c) array1[1]
- d) array1[0]**

33. AVL Tree is?

a) Non Linear data structure

- b) Linear data structure
- c) Hybrid data structure (Mixture of Linear and Non Linear)
- d) None of the given options.

34. The operation for removing an entry from a stack is traditionally called:

- a) delete
- b) peek
- c) pop**
- d) remove

35. Which statement of the following statements is incorrect?

- a) Lists can be implemented by using arrays or linked lists
- b) A list is a sequence of one or more data items
- c) Stack is a special kind of list in which all insertions and deletions take place at one end**
- d) Stacks are easier to implement than lists

36. Which one of the following statement is NOT correct?

- a) In linked list the elements are necessarily to be contiguous**
- b) In linked list the elements may locate at far positions in the memory
- c) In linked list each element also has the next to it
- d) In an array the elements are contiguous

37. Parameters in function call are passed using?

- a) Stack**
- b) Queue
- c) Binary Search Tree
- d) AVL Tree

38. Queue follows?

- a) Last in First out
- b) First in Last out
- c) First in First out**
- d) None of these

39. _____ is a data structure that can grow easily dynamically at run time without having to copy existing elements?

- a) Array
- b) List**
- c) Both of these
- d) None of these

40. Which one of the following statement is NOT correct?

- a) In linked list the elements are necessarily to be contiguous
- b) In linked list the elements may locate at far positions in the memory**
- c) In linked list each element also has the next to it
- d) In an array the elements are contiguous

41. The binary operator (+, -, /, *) are evaluated in _____ direction?

- a) Left-to-right**
- b) Right-to-left
- c) In both direction
- d) All of the above

42. What is the size of following array?

`char str[] + "abcd"`

a) 5

b) 4

c) 2

d) 3

43. Each operator in a postfix expression refers to the previous _____ operand(s).

a) One

b) Two

c) Three

d) Four

44. Suppose `currentNode` refers to a node in a linked list (using the `Node` class with member variables called `data` and `nextNode`). What statement changes `currentNode` so that it refers to the next node?

a) `currentNode ++;`

b) `currentNode = nextNode;`

c) `currentNode += nextNode;`

d) `currentNode = currentNode->nextNode;`

45. A queue where the de-queue operation depends not on FIFO, is called a priority queue?

a) False

b) True

46. I have implemented the queue with a linked list, keeping track of a front pointer and a rear pointer. Which of these pointers will change during an insertion into an EMPTY queue?

a) Neither changes

- b) Only front pointer changes.
- c) Only rear pointer changes.**
- d) Both change.

47. The nodes with no successor are called _____?

- a) Root Nodes
- b) Leaf Nodes**
- c) Both of these
- d) None of these

48. A binary search tree should have minimum of one _____ node/s at each level?

- a) One
- b) Two**
- c) Three
- d) Four

49. In the statement `int x[6];`, we cannot assign any value to `x` because `x` is not an value?

- a) True
- b) False**

50. A tree is an AVL tree if

- a) Any one node fulfills the AVL condition
- b) At least half of the nodes fulfill the AVL condition
- c) All the nodes fulfill the AVL condition**
- d) None of the given options

51. What will be postfix expression of the following infix expression?

Infix Expression: $a+b*c-d$

- a) $ab+c*d-$
- b) $abc*+d-$**
- c) $abc+*d-$
- d) $abcd+*-$

52. Which one of the following calling methods does not change the original value of the argument in the calling function?

- a) None of the given options
- b) Call by passing the value of the argument**
- c) Call by passing reference of the argument
- d) Call by passing the address of the argument

53. In a program a reference variable, say x, can be declared as

- a) $\text{int } \&x ;$**
- b) $\text{int } *x ;$
- c) $\text{int } x ;$
- d) None of the given options

54. Consider the following pseudo code declare a stack of characters

while (there are more characters in the word to read)

{

read a character

push the character on the stack

}

while (the stack is not empty)

```
{  
pop a character off the stack  
write the character to the screen  
}
```

What is written to the screen for the input "apples"?

- a) selpa
- b) selppa**
- c) apples
- d) aaappppplleess

55. In the following C++ code, how many function calls are made?

```
int x, y, z;  
x = 2;  
y = 3 + x;  
z = foobar(x,y);
```

- a) 1
- b) 4
- c) 7
- d) 8**

56. Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data

and nextNode). What statement changes currentNode so that it refers to the next node?

- a) currentNode ++;
- b) currentNode = nextNode;
- c) currentNode += nextNode;
- d) currentNode = currentNode->nextNode;**

57. A queue where the de-queue operation depends not on FIFO, is called a priority queue?

a) False

b) True

58. We can add elements in QUEUE From _____?

a) Front

b) Rear

c) From Both Rare and Front

d) None of these

59. How many of the nodes have at least one sibling?

a) 8

b) 7

c) 5

d) 6

60. The _____ method of list data structure removes the element residing at the current position?

a) Add

b) next

c) remove

d) find

61. It will be efficient to place stack elements at the start of the list because insertion and removal take _____ time?

a) Variable

b) Constant

- c) Inconsistent
- d) None of the above

62. The nodes with no successor are called _____?

- a) Root Nodes
- b) Leaf Nodes**
- c) Both of these
- d) None of these

63. In which of the traversal method, the recursive calls can be used to traverse a binary tree?

- a) In preorder traversal only**
- b) In inorder traversal only
- c) In postorder traversal only
- d) All of the given option

64. Addition of new items in stack make the pointer ----- by 2?

- a) Increment, bytes
- b) Increment, bits
- c) Decrement, bits
- d) Decrement, bytes**

65. Next item in a linked list is known as?

- a) Index
- b) Item
- c) Node**
- d) Child

66. In an AVL tree to delete a parent with two childs in a straight line following rotations will be required?

- a) Single
- b) Double**
- c) Triple
- d) None

67. To check the depth of?

- a) $1.66 \log_2 n$
- b) $1.44 \log_2 n$**
- c) $\log_2 (n+1) - 1$
- d) $1.66 \log_2 n (n+1)$

68. Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What boolean expression will be true when cursor refers to the tail node of the list?

- a) (currentNode == null)**
- b) (currentNode->nextNode == null)
- c) (nextNode.data == null)
- d) (currentNode.data == 0.0)

69. Which of the following tests in the client code correctly compares two class objects alpha and beta?

- a) if (alpha < beta)
- b) if (alpha.LessThan(beta))**
- c) if (LessThan(alpha, beta))
- d) if (LessThan(alpha).beta)

70. BST is a _____ Structure.

- a) Linear
- b) Non Linear**
- c) Circular
- d) None of Above

71. After creation of an array?

- a) Size can be increase but can not be decreased.
- b) Size can be decreased but can not be increased.
- c) Size can neither be increased nor be decreased.**
- d) Size can be increased and can also be decreased.

72. Following are the linear data structures?

- a) Stacks
- b) Queue
- c) Both a & b**
- d) None of the above

73. In C what is the operation that you can not do with primitive types?

- a) Assign a value to primitive type using a literal
- b) Declare primitive types to be constant using the Const keyword
- c) Create a new instance of primitive type with New keyword**
- d) None of these

74. When an operator is used in between two operands this is which type of Notation?

- a) Prefix
- b) Postfix
- c) Infix**
- d) None of the Above

75. Suppose that the class declaration of SomeClass includes the following function prototype.

`bool LessThan( SomeClass anotherObject );`

Which of the following tests in the client code correctly compares two class objects alpha and beta?

- a) `if (alpha < beta)`
- b) `if (alpha.LessThan(beta))`**
- c) `if (LessThan(alpha, beta))`
- d) `if (LessThan(alpha).beta)`

76. Which of the following can be used to reverse a string value?

- a) Stack**
- b) Queue
- c) Both of these
- d) None of these

77. The following are statements related to queues.

- (i) The last item to be added to a queue is the first item to be removed
- (ii) A queue is a structure in which both ends are not used
- (iii) The last element hasn't to wait until all elements preceding it on the queue are removed
- (iv) A queue is said to be a last-in-first-out list or LIFO data structure.

Which of the above is/are related to normal queues?

- a) (iii) and (ii) only
- b) (i), (ii) and (iv) only
- c) (ii) and (iv) only
- d) None of the given options**

78. An array is a group of consecutive related memory locations.

a) True

b) False

79. In an array we can store data elements of different types?

a) True

b) False

80. Which one of the following statements is NOT correct?

a) Array size can be changed after its creation.

b) Link List size can be changed after its creation

c) Binary Search Tree size can be changed after its creation

d) AVL Tree size can be changed after its creation

81. A queue is a----- data structure, whereas a stack is a -----data structure.

a) FIFO, LIFO

b) LIFO,FIFO

c) none of these

d) both of these

82. Which one of the following operators has higher priority than all of others?

a) Multiplication operator

b) Minus operator

c) Plus operator

d) Exponentiation operator

83. The operation for adding an entry to a stack is traditionally called?

- a) add
- b) append
- c) insert**
- d) Push

84. Consider the following sequence of push operations in a stack:

stack.push('7');

stack.push('8');

stack.push('9');

stack.push('10');

stack.push('11');

stack.push('12');

- a) 7 8 9 10 11 12**
- b) 9 8 11 10 7 12
- c) 9 10 8 11 12 7
- d) 9 10 8 12 7 11

85. The following are statements related to queues.

- (i) The last item to be added to a queue is the first item to be removed
- (ii) A queue is a structure in which both ends are not used
- (iii) The last element hasn't to wait until all elements preceding it on the queue are removed
- (iv) A queue is said to be a last-in-first-out list or LIFO data structure.

Which of the above is/are related to normal queues?

- a) (iii) and (ii) only**
- b) (i), (ii) and (iv) only
- c) (ii) and (iv) only

- d) None of the giv
- e)
- f)
- g) en options

86. Four statements about trees are below. Three of them are correct. Which one is INCORRECT?

- a) Trees are recursively defined multi-dimensional data structures tree
- b) The order of a tree indicates a maximum number of children allowed at each node of the
- c) A search tree is a special type of tree where all values (i.e. keys) are ordered

d) If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height.

87. Which of the following is "TRUE" about arrays?

- a) We can increase the size of arrays after their creation.
- b) We can decrease the size of arrays after their creation.
- c) We can increase but can't decrease the size of arrays after their creation.

d) We can neither increase nor decrease the array size after their creation.

88. Consider the following statements.

(i) A binary tree can contain at least $2L$ Nodes at level L .

(ii) A complete binary tree of depth d is a binary tree that contains $2L$ Nodes at each level L

between 0 and d , both inclusive.

(iii) The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.

(iv) The height of the complete binary tree can be written as $h = \log_2 (T_n + 1) - 1$ where T_n is Total number of Nodes.

Which one of the following is correct in respect of the above statements regarding the Binary trees?

- a) (i) and (iii) only
- b) (i), (ii) and (iii) only
- c) (ii) and (iii) only
- d) (ii), (iii) and (iv) only**

89. Consider the following infix expression.

$$5 + 6/2$$

If one converts the above expression into postfix, what would be the resultant expression?

- a) $56/ + 2$
- b) $5\ 6\ 2\ /\ +$**
- c) $5\ 6\ /\ 2\ +$
- d) $/62 + 5$

90. There is/are _____ case/s for rotation in an AVL tree?

- a) 1**
- b) 3
- c) 2
- d) 4

91. In an AVL tree to delete a parent with two childs in a straight line following rotations will be required?

- a) Single
- b) Double**
- c) Triple
- d) None.

92. Which one of the following is used to remove the item from of the queue?

- a) pop()
- b) remove()
- c) dequeue()**
- d) All of the given option

93. The memory address of the first element of an array is called _____?

- a) Floor address
- b) Foundation address
- c) Base address**
- d) First element

94. What will the result of the following athematic expression?

$$6 + 27 / 3 * 3$$

- a) 33
- b) 45
- c) 9**
- d) 30

95. Which of the following function is used to get the length of the List?

- a) start()
- b) tail()
- c) length()**
- d) get()

96. The operation for adding an entry to a stack is traditionally called?

- a) add

- b) append
- c) insert
- d) push**

97. _____ is the maximum number of nodes that you can have on a stack-linked list?

- a) Zero
- b) $2n$
- c) Any number**
- d) None of the these

98. Highest Operators Precedence is of the following operator?

- a) Plus
- b) Minus
- c) Multiply**
- d) Exponentiation

99. A Compound Data Structure is the data structure which can have multiple data items of same type or of different types. Which of the following can be considered compound data structure?

- a) Arrays**
- b) LinkLists
- c) Binary Search Trees
- d) All of the given options

100. Suppose a pointer has been declared in main but has not assigned any variable address then?

- a) That pointer points to First byte in main function
- b) That pointer contains a NULL value
- c) None of these

d) That pointer points to any memory address

101. Which of the following operation return the left sub tree in Binary search tree?

- a) Parent()
- b) Right()
- c) Left()**
- d) All of the Above

102. A binary search tree should have minimum of one _____ node/s at each level?

- a) 1**
- b) 2
- c) 3
- d) 4

103. We can add elements in QUEUE From _____?

- a) Front
- b) Rear**
- c) From Both Rare and Front
- d) None of these

104. Which of the following abstract data types are NOT used by Integer Abstract Data type group?

- a) short
- b) Int
- c) float**
- d) long

105. From which function we more to get the list item?

- a) start()
- b) tail()
- c) length()
- d) get()**

106. Which of the following is not in data structure?

- a) List
- b) Linked list
- c) Circular List
- d) None of the above**

107. Consider the following function: void test_a(int n) { cout << n << " "; if (n>0) test_a(n-2); } What is printed by the call test_a(4)?

- a) 4 2 0**
- b) 0 2 4
- c) 0 2
- d) 2 4

108. Which statement of the following statements is incorrect?

- a) Lists can be implemented by using arrays or linked lists
- b) A list is a sequence of one or more data items
- c) Stack is a special kind of list in which all insertions and deletions take place at one end
- d) Stacks are easier to implement than lists**

109. Each entry which points to a null value in a Singly Linked List is known as?

- a) Node
- b) First Node
- c) Last Node**
- d) Head Node

110. Each node in a BST has Pointers?

- a) 1
- b) 2**
- c) 3
- d) 4

111. To check the depth of an AVL tree following time will be taken?

- a) $1.66 \log_2 n$
- b) $1.44 \log_2 n$**
- c) $\log_2 (n+1) - 1$
- d) $1.66 \log_2 n (n+1)$

112. Each operator in postfix expression refers to the previous _____ operand(s)?

- a) One
- b) Two**
- c) Three
- d) Four

113. We access the elements in AVL Tree in?

- a) Linear way
- b) Non linear way**
- c) Both a and b
- d) None of the above

114. Which of the following function used to connect the next node with the current node?

- a) next()
- b) getNext()

c) setNext()

d) get()

115. ADT stand for _____?

a) Advance Data Type

b) Abstract Data Type

c) Alternative Data Type

d) All of the above

116. What will be the postfix notation of $5+6/2$?

a) $56+/2$

b) $562+ /$

c) $5+62/$

d) $562/$

117. For a perfect binary tree of height 4. What will be the sum of heights of nodes?

a) 31

b) 30

c) 27

d) 26

118. Consider this function declaration:

```
void quiz(int i)
{
if (i > 1)
{
quiz(i / 2);
quiz(i / 2);
}
```

```
cout << "*";  
}
```

How many asterisks are printed by the function call *quiz(5)*?

- a. 3
- b. 4
- c. 7**
- d. 8

119. Which one of the following statement is NOT correct?

- a) In linked list the elements are necessarily to be contiguous
- b) In linked list the elements may locate at far positions in the memory**
- c) In linked list each element also has the next to it
- d) In an array the elements are contiguous

120. Suppose T is a binary tree with 14 nodes. What is the minimum possible depth of T?

- a) 0
- b) 3**
- c) 4
- d) 5

121. Which one of the following calling methods does not change the original value of the argument in the calling function?

- a) None of the given options
- b) Call by passing the value of the argument**
- c) Call by passing reference of the argument
- d) Call by passing the address of the argument

122. A tree is an AVL tree if

- a) Any one node fulfills the AVL condition
- b) At least half of the nodes fulfill the AVL condition
- c) All the nodes fulfill the AVL condition
- d) None of the given options**

123. "Entries in a stack are Ordered". What is the meaning of this statement?

- a) A collection of stacks can be sorted.
- b) Stack entries may be compared with the < operation.
- c) The entries must be stored in a linked list.
- d) There is a first entry, a second entry, and so on.

124. Suppose `currentNode` refers to a node in a linked list (using the `Node` class with member variables called `data` and `nextNode`). What statement changes `currentNode` so that it refers to the next node?

- a) `currentNode ++;`
- b) `currentNode = nextNode;`
- c) `currentNode += nextNode;`
- d) `currentNode = currentNode->nextNode;`

125. Which of the following applications may use a stack?

- a) A parentheses balancing program.
- b) Keeping track of local variables at run time.
- c) In-order traversal of a binary tree.
- d) All of the above.

126. A queue where the de-queue operation depends not on FIFO, is called a priority queue.

- a) False
- b) True

127. Which one is a self- referential data type?

- a) Stack
- b) Queue
- c) Link list
- d) All of these

128. In which location do dynamic variables reside?

- a) The code segment.
- b) The data segment.
- c) The heap.**
- d) The run-time stack.

129. Select the one FALSE statement about binary trees:

- a) Every binary tree has at least one node.
- b) Every non-empty tree has exactly one root node.**
- c) Every node has at most two children.
- d) Every non-root node has exactly one parent.

130. Select the one FALSE statement about binary trees:

- a) Every binary tree has at least one node.
- b) Every non-empty tree has exactly one root node.**
- c) Every node has at most two children.
- d) Every non-root node has exactly one parent.

131. Which one of the following statement is NOT correct.

- a) In linked list the elements are necessarily to be contiguous
- b) In linked list the elements may locate at far positions in the memory**
- c) In linked list each element also has the next to it
- d) In an array the elements are contiguous

132. A binary search tree should have minimum of one _____ node/s at each level.

- a) One**
- b) Two
- c) Three
- d) Four

133. In the call by value methodology, a copy of the object is passed to the called function.

a) False

b) True

134. When should you use a const reference parameter?

a) Whenever the parameter has huge size

b) Whenever the parameter has huge size, the function changes the parameter within its body, and you do NOT want these changes to alter the actual argument.

c) Whenever the parameter has huge size, the function changes the parameter within its body, and you DO want these changes to alter the actual argument.

d) Whenever the parameter has huge size, and the function does not change the parameter within its body.

135. Here is the start of a C++ class declaration:

```
class foo
```

```
{
```

```
public:
```

```
void x(foo f);
```

```
void y(const foo f);
```

```
void z(foo f) const;
```

```
...
```

Which of the three member functions can alter the PRIVATE member variables of the foo object that activates the function?

a) Only x can alter the private member variables of the object that activates the function.

b) Only y can alter the private member variables of the object that activates the function.

c) Only z can alter the private member variables of the object that activates the function.

d) Two of the functions can alter the private member variables of the object that activates the function.

136. In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?

- a) At the head
- b) At the tail
- c) After all other entries that are greater than the new entry.
- d) After all other entries that are smaller than the new entry.

137. Suppose we have a *circular* array implementation of the queue class, with ten items in the queue stored at data[2] through data[11]. The CAPACITY is 42, i.e., the array has been declared to be of size 42. Where does the push member function place the new entry in the array?

- a) data[1]
- b) data[2]
- c) data[11]
- d) data[12]

138. Suppose a pointer has been declared in main but has not assigned any variable address then

- a) That pointer points to First byte in main function
- b) That pointer contains a NULL value
- c) None of these
- d) That pointer points to any memory address

139. Which statement of the following statements is incorrect?

- a) Lists can be implemented by using arrays or linked lists
- b) A list is a sequence of one or more data items
- c) Stack is a special kind of list in which all insertions and deletions take place at one end
- d) Stacks are easier to implement than lists

140. Function signatures are also called_____

a) Function Prototype

- b) Function Overloading
- c) Function Definition
- d) Function Overriding

141. The next field in the last node of the singly linked list is set to_____

- a) NAN
- b) 1
- c) NULL**
- d) -1

142. The main use of AVL tree is_____

- a) Searching of data
- b) Storing of data**
- c) Insertion of data
- d) Security of data

143. In simple implementation of stack, isFull() method is used due to_____

- a) Limitation of array**
- b) Strength of array
- c) Linked list of connectivity
- d) Complexity of linked list

144. For a perfect binary tree of height h, having N nodes the sum of height of nodes is_____

- a) N-h-1**
- b) N-1
- c) N-1+h
- d) -(h-1)

145. Sorting procedure normally takes _____time

a) $N \log N$

b) $2N$

c) $N*N*N$

d) N

146. there are 100 elements in an equivalence classes then will have _____state initially

a) 50

b) 100

c) 1000

d) 80

147. For making binary search tree for strings we need, _____ data type.

a) Char

b) Int

c) Float

d) Double

148. Linked list use _____ to store data.

a) Array

b) 2-D Array

c) Variables

d) Linked Memory

149. To represent hierarchical relationship between elements, which data structure is suitable?

a) Dequeue

b) Priority

c) Stack

d) Tree

150. In-order traversal method traverses the data in_____

a) Non sorted order

b) Random order

c) Sorted order

d) None of the given

151. Longest path from the root node to farthest leaf node is called_____

a) Level

b) Length

c) Depth

d) Node level

152. The_____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

a) Height

b) Balance

c) Width

d) None of the above

153. Which statement of the following statements is incorrect?

a) Lists can be implemented by using arrays or linked lists

b) A list is a sequence of one or more data items

c) Stack is a special kind of list in which all insertions and deletions take place at one end

d) Stacks are easier to implement than lists

154. _____ is a self-balancing tree.

a) AVL

- b) Binary Tree
- c) Binary Search Tree
- d) ALV

155. `Int htdiff = height(root->getLeft()) _____`

`height(root->getRight());` The above line of code is taken from AVL insert method.

Complete it by selecting an appropriate symbol.

a) -

- b) +
- c) /
- d) *

156. Local variables defined inside function body are _____ automatically at the end of function execution.

- a) Created
- b) Destroyed**
- c) Incremented
- d) Decrementd

157. Which operation of the queue data structure is used to insert an element into the Queue?

a) Enqueue()

- b) Dequeue()
- c) Fornt()
- d) Remove()

158. While joining nodes in the building of Huffman encoding tree if there are more

nodes with same frequency, we choose the nodes _____.

- a) Randomly**
- b) That occur first in the text message
- c) That are lexically smaller among others.
- d) That are lexically greater among others

159. In-order traversal method traverses the data in_____

- a) Non sorted order
- b) Random order
- c) Sorted order**
- d) None of the given

160. In a tree, we link the nodes in such a way that it _____ a linear structure.

- a) Does not remain**
- b) Forms
- c) Reverses
- d) Remains

161. Consider the following paragraph with blanks.

A is a linear list where and take place at the same end . This end is called the

What would be the correct filling the above blank positions?

- a) (i) queue (ii) insertion (iii) removals (iv) top
- b) (i) stack (ii) insertion (iii) removals (iv) bottom
- c) (i) stack (ii) insertion (iii) removals (iv) top**
- d) (i) tree (ii) insertion (iii) removals (iv) top

162. Binary search tree violates the condition of AVL tree when any node has balance equal to_____

- a) 2 or -2
- b) 1 or -1**

- c) 0
- d) None of the options

163. Security of data is the main usage of AVL tree.

- a) True
- b) False**

164. In internal memory organization of a process, there is some area of memory for static data that holds _____ variables.

- a) Static
- b) Global
- c) Not static neither global
- d) Both static and global**

165. If we use doubly linked list to implement list then there is an issue of

- a) Next pointer of first node and pre pointer of last node are NULL
- b) Next pointer of first node and next pointer of last node are NULL
- c) Pre pointer of first node and next pointer of last node are NULL**
- d) Pre pointer of first node and pre pointer of last node are NULL

166. Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function.

- a) First argument
- b) Return address**
- c) Last argument
- d) None of the above

167. Suppose a stack class has been defined using template. Now, we want to declare a Stack object of an int type. What will be the correct syntax?

- a) <int>Stack stack;
- b) Stack<int> stack;**
- c) Stack int stack;
- d) Int Stack stack;

168. A binary tree with 33 internal nodes has _____ links to internal nodes.

- a) 31
- b) 32**
- c) 33
- d) 66

169. ~BinarySearchTree() is a _____.

- a) Constructor
- b) Destructor**
- c) Switch case
- d) Template method call

170. Which traversal gives a decreasing order of elements in a heap where the max element is stored at the top?

- a) post-order**
- b) level-order
- c) inorder
- d) None of the given options

171. How many cases of rotation are there in AVL tree?

- a) 2
- b) 4**
- c) 6
- d) 8

172. `Int * i = new int[10];`

Above given code will:

- a) Create an integer having value 10
- b) Allocate memory for 9 integers
- c) Allocate memory for 10 integers**
- d) Create 10 pointers of integer type

173. Which of the following is a non linear data structure?

- a) Linked List
- b) Stack
- c) Queue
- d) Tree**

174. Allocating and de-allocating memory for linked list nodes

does take _____ time than pre-allocated array.

- a) Less
- b) More**
- c) Equal
- d) No

175. In the statement `int x[6];`, we cannot assign any value to x because x is not an lvalue.

- a) True
- b) False**

176. For reference variables, _____ sign is used.

- a) Ampersand**
- b) Asterisk
- c) Sigma
- d) Dollar

177. In a program a reference variable, say x, can be declared as

- a) **int &x ;**
- b) int *x ;
- c) int x ;
- d) None of the given options

178. The - - is a decrement operator in C++ that decreases the value of the operand by ____.

- a) **One**
- b) Two
- c) Three
- d) Four

179. int x, y, z;

x = 2;

y = 3 + x;

z = foobar(x,y);

- a) 1
- b) 4
- c) 7
- d) **8**

180. When an executable program runs, it is loaded in the computer memory and becomes a ____.

- a) Thread
- b) .h file
- c) **Process**
- d) None of the above

181. Get(?) method of list class is used to:

- a) Get element from the last position
- b) Get element from the first position
- c) Get element from the middle position
- d) Get element at the given position**

182. We can add elements in QUEUE From _____

- a) Front**
- b) Rear
- c) From Both Rare and Front
- d) None of these

183. STL is a _____ that is a part of the official standard of C++.

- a) C program file
- b) .h file
- c) .cpp file
- d) Library**

184. How many descendants does the root have?

- a) 5
- b) 6
- c) 7
- d) 8**

185. _____ is when function is calling to itself.

- a) Loop
- b) Recursion**
- c) Iteration
- d) Nested loop

186. Below is a binary search tree. If we delete the value 50 using the algorithm we discussed, what

value will be in the root of the remaining tree?

- a) 50
- b) 60
- c) 70**
- d) 80

187. Following is a keyword of C++

- a) Del
- b) Delete**
- c) Remove
- d) Eliminate

188. Which of the following statement regarding binary tree is NOT correct.

- a) A binary tree can contain at least $2L$ Nodes at level L .
- b) A complete binary tree of depth d is a binary tree that contains $2L$ Nodes at each level L between 0 and d , both inclusive.
- c) The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.
- d) The height of the complete binary tree can be written as $h = \log_2 (T_n + 1) - 1$ where T_n is Total number of Nodes.**

189. A software solution is said to be efficient if it solves the problem_____.

- a) By using some extra resources
- b) Within no time
- c) By consuming more hardware resources
- d) Within its resources constraints**

190. The following are statements related to queues.

- i. The last item to be added to a queue is the first item to be removed
- ii. A queue is a structure in which both ends are not used

- iii. The last element hasn't to wait until all elements preceding it on the queue are removed
iv. A queue is said to be a last-in-first-out list or LIFO data structure.

Which of the above is/are related to normal queues?

- a) (iii) and (ii) only**
- b) (i), (ii) and (iv) only
- c) (ii) and (iv) only
- d) None of the given options

191. add(12) method of linked list class will

- a) Add 12 nodes in linked list
- b) Add 12 pointers in linked list
- c) Add 12 as value in linked list**
- d) Add 12 values in linked list

192. The _____ method of list data structure removes the element residing at the current position.

- a) Add
- b) next
- c) remove**
- d) find

193. back() method of list class is used to _____

- a) Moves the "current" pointer to backward one element.**
- b) Moves the "current" pointer to backward two element.
- c) Moves the "current" pointer to backward three element.
- d) Moves the "current" pointer to backward four element.

194. It will be efficient to place stack elements at the start of the list because insertion and removal take _____ time.

- a) Variable
- b) Constant**

- c) Inconsistent
- d) None of the above

195. A queue is a data structure where elements are

- a) Inserted at the front and removed from the back
- b) Inserted and removed from the top
- c) Inserted at the back and removed from the front
- d) Inserted and removed from both ends**

196. Which one of the following statement is NOT correct.

- a) In linked list the elements are necessarily to be contiguous
- b) In linked list the elements may locate at far positions in the memory**
- c) In linked list each element also has the address of the element next to it
- d) In an array the elements are contiguous

197. An efficient program execute faster and helps in _____ the usage of resources like memory and disk.

- a) Maximizing
- b) Minimizing**
- c) Equalizing
- d) None of the given

198. Which of the following abstract data types are NOT used by Integer Abstract Data type group?

- a) short
- b) int
- c) float**
- d) long

199. Generalized code written for a class is called:

- a) Function
- b) Template**
- c) Structure
- d) Stack

200. The operation for adding an entry to a stack is traditionally called_____

- a) add
- b) append
- c) insert
- d) push**

201. In C++, we place the class interface in ____ file.

- a) .cpp
- b) .cppp
- c) .h**
- d) .hh

202. The operation for removing an entry from a stack is traditionally called:

- a) delete
- b) peek
- c) pop**
- d) remove

203. Which of the following function don't belongs to the stack class?

- a) push()
- b) pop()
- c) crash()**
- d) top()

204. The difference between a binary tree and a binary search tree is that ,

- a) a binary search tree has two children per node whereas a binary tree can have none, one, or two children per node
- b) in binary search tree nodes are inserted based on the values they contain**
- c) in binary tree nodes are inserted based on the values they contain
- d) none of these

205. The back() method decreases the value of Variable current by_____

- a) Four
- b) Three
- c) Two
- d) One**

206. The following is a segment of a C program.

```
int pqr(BinaryNode t)
{ if (t == null )
return -1;
18
else
return 1+max(pqr(t.left),pqr(t.right)) }
```

Identify, what the above program intend(s) to do?

- a) Compute the height of a binary tree using an in-order traversal
- b) Compute the height of a binary tree using a pre-order traversal
- c) Compute the depth of a binary tree using a pre-order traversal
- d) Compute the depth of a binary tree using a post-order traversal**

207. Suppose we have the following values to be inserted in constructing AVL tree,

10, 13, 15, 5, 7, 8

Tell when first rotation will take place.

- a) After inserting node 13**
- b) After inserting node 15
- c) After inserting node 5
- d) After inserting node 7

208. Consider the following infix expression:

$$3 + 5 * 6 - 7 * (8 + 5)$$

Which of the following is a correct equivalent expression(s) for the above?

- a) $3\ 6\ 5 + * 7\ 5\ 8 + - *$
- b) $3\ 6\ 5\ 7\ 5\ 8 + * + - *$
- c) $3\ 5\ 6 + * 7\ 8\ 5 + - *$
- d) $3\ 5\ 6 * + 7\ 8\ 5 + * -$

209. Binary search algorithm cannot be applied to ____

- a) **Sorted linked list**
- b) sorted binary trees
- c) sorted linear array
- d) None of given option

210. An array is a group of consecutive related memory locations.

- a) **True**
- b) False

211. ____ is the major factor to see the efficiency of a program.

- a) Quality
- b) **Time**
- c) Correctness
- d) None of the given

212. Deleting a leaf node in binary search tree involves setting ____ pointer/s of that nodes parent as null.

- a) **1**
- b) 2
- c) 3

d) 4

213. A template is a function or class that is written with a _____

- a) Specific
- b) Definite
- c) Generic**
- d) None of the above

214. subscript of an array may be an integer or an integer expression.

- a) True**
- b) False

215. In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?

- a) After all other entries that are greater than the new entry
- b) At the head**
- c) After all other entries that are smaller than the new entry
- d) At the tail

216. Doubly Linked List always has one NULL pointer.

- a) True
- b) False**

217. Linked list is generally considered an example of _____ type of memory location.

- a) Static
- b) Compile time

c) Dynamic

d) None of the above

218. suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What boolean expression will be true when cursor refers to the tail node of the list?

a) (currentNode == null)

b) (currentNode->nextNode == null)

c) (nextNode.data == null)

d) (currentNode.data == 0.0)

219. copy() method of list data structure _____.

a) copy first item of list

b) set one list to be a copy of another

c) copy last item of list

d) copy any item of list 220

221. NULL is an invalid address and _____.

a) Accessible

b) Inaccessible

c) Points to the start point of the list

d) Points to the last point of the list

222. Suppose that the class declaration of SomeClass includes the following function prototype.

bool LessThan(SomeClass anotherObject);

Which of the following tests in the client code correctly compares two class objects alpha and beta?

a) if (alpha < beta)

b) if (alpha.LessThan(beta))

- c) if (LessThan(alpha, beta))
- d) if (LessThan(alpha).beta)

223. In a complete binary tree, for 25000 nodes the depth will be_____.

- a) 13
- b) 14**
- c) 15
- d) 16

224. if there are ____ nodes in an avl tree its levels will be roughly as $\log_2(10 \text{ million})$.

- a) 100 million
- b) 10 million**
- c) 5 million
- d) 2 million

225. The smallest value element in a binary search tree(Each node with left and right pointer) lies at

- a) Root Node
- b) Left Child of Root
- c) Right Most Node
- d) Left Most Node**

226. In C what is the operation that you cannot do with primitive types?

- a) Assign a value to primitive type using a literal
- b) Declare primitive types to be constant using the Const keyword
- c) Create a new instance of primitive type with New keyword**
- d) None of these

227. Which one is the correct function call for the following function

of calculatincube?

```
int cube(int&num)
```

```
{
```

```
·
```

```
·
```

```
·
```

- a) Cube(&num)
- b) Cube(&&num)
- c) Cube(*num)
- d) Cube(num)**

228. For String-based Binary Search Tree, we use ASCII values of characters for comparing among letters. This method is known as____

- a) Lexicographic order**
- b) Alphabet coding procedure
- c) Asymmetric technique
- d) heap-based approach

229. _____ is the maximum number of nodes that you can have on a stack-linked list?

- a) Zero
- b) $2n$ (where n is the number of nodes in linked list)
- c) Any Number**
- d) None of these

230. Elements in a queue data structure are added from____ and remove from____ .

- a) Rear end. front end**
- b) Front end. Rear end
- c) Front end

d) rear end

231. In an array list the current element is_____

- a) The first element
- b) The middle element
- c) The last element
- d) The element where the current pointer points to**

232. If numbers 5,222,4,48 are inserted in a queue. Which one will be remove first?

- a) 48**
- b) 4
- c) 222
- d) 5

233. which one of the following operators has higher priority than all of others?

- a) Multiplication operator
- b) Minus operator
- c) Plus operator
- d) Exponentiation operator**

234. Four statements about trees are below. Three of them are correct. Which one is INCORRECT?

- a) Trees are recursively defined multi-dimensional data structures
- b) The order of a tree indicates a maximum number of children allowed at each node of the tree
- c) A search tree is a special type of tree where all values (i.e. keys) are ordered
- d) If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height.**

235. `y = &x[0];`

In the above statement, we get the address of the first location of the array `x` and store it in `y`.

Here “`y`” is:

- a) rvalue
- b) xvalue
- c) lvalue**
- d) zvalue

236. In the following C++ code, how many function calls are made?

```
int x, y, z;  
x = 2;  
y = 3 + x;  
z = foobar(x,y);
```

- a) 1**
- b) 4
- c) 7
- d) 8

237. Suppose we have the following values to be inserted in constructing AVL tree,

20,23,25,10, 12,13

Tell when first rotation will take place.

- a) After increasing Node 25**
- b) After increasing Node 23
- c) After increasing Node 10
- d) After increasing Node 12

238. The percolate down procedure will move the smaller value___and Bigger value___

- a) Left, right
- b) Right, left
- c) Down, up
- d) Up, down**

239. Addition of new items in stack make the pointer ----- by 2.

- a) Increment, bits
- b) Increment, bytes
- c) Decrement, bits
- d) Decrement, bytes**

240. if (! heap ->isEmpty()) checks

- a) Heap is empty
- b) Heap is full
- c) Heap is not empty**
- d) Not a valid expression

241. Which of the is NOT true regarding the maze generation?

- a) Randomly remove walls until the entrance and exit cells are in same set.
- b) Removing a wall is the same as doing a union operation
- c) Remove a randomly chosen wall if the cells it separates are already in same set**
- d) Do not remove a randomly chosen wall if the cells it separates are already in same set.

242. Which one of the following operations returns top value of the stack?

- a) Push
- b) Pop
- c) Top**
- d) First

243. If Ahmad is cousin of Ali and Ali is cousin of Asad then Ahmad is also cousin of Asad. This statement has the following property

- a) Reflexivity
- b) Symmetry
- c) Transitivity
- d) All of the given

244. The array in binary search is sub divided _____.

- a) Once
- b) Twice
- c) N time
- d) Until a sublist is no more divisible

245. Here is a small function definition:

```
void f(int i, int &k)
```

```
k = 2;
```

```
}
```

Suppose that a main program has two integer variables x and y, which are given the value 0. Then the main program calls f(x,y); What are the values of x and y after the function f finishes?

- a) Both x and y are still 0.
- b) x is now 1, but y is still 0.
- c) x is still 0, but y is now 2.
- d) x is now 1, and y is now 2.

246. What is the hash function used in linear probing?

- a) $h_i(x) = \text{hash}(x) \bmod \text{table size}$
- b) $h_i(x) = (\text{hash}(x) + f(i^2)) \bmod \text{table size}$
- c) $h_i(x) = (\text{hash}(x) + f(i)) \bmod \text{table size}$

d) $hi(x) = X \bmod 17$

247. In singly linked list a node consists of two parts:

- a) Object and structure
- b) Two pointers
- c) Two objects
- d) Object and pointer**

248. Before using the Pop method of a stack, the user must call the _____ method.

a) isFull()

- b) push()
- c) pop()
- d) isEmpty()

249. “end()” method of list performs its tasks in

- a) Many Steps
- b) Three Steps
- c) One Steps**
- d) Two Steps

250. Complete the push method code of stack `void push (int x) { A{____}=x}`

- a) `Count++`
- b) `Count—`
- c) `++count`**
- d) `—count`

251. Stack push(15) will push 15 on _____

a) Top of the Stack

- b) Anywhere of the Stack
- c) Middle of the Stack
- d) Bottom of the Stack

252. Last Node in Circular Linked list Contains.

- a) Maximum Two points
- b) No Null Pointer**
- c) At least one Null pointer
- d) None

253. Analyze the given code carefully and identify which type of binary tree traversal this is;

```
Void traversal ( treeNode*treeNode)
```

```
{ if(treeNode !=Null)
```

```
{ cout << *(treeNode->getinfo( ))<< “ ”; Traversal ( treeNode-
```

```
>getleft( )); Traversal ( treeNode->getRight( )); } }
```

- a) Inorder
- b) Preorder**
- c) Postorder
- d) Sorted Order

254. We can make a lexicographic order of characters based on their_____.

- a) RANDOM choice
- b) ASCII values**
- c) Binary DIGITS
- d) Memory Address

255. If we return the reference of a local variable from a function it will cause_____

- a) Reference Overloading

- b) Duplication of local variables
- c) Deletion of local variables from memory
- d) Dangling reference**

256. If a function has recursive call as the last statement, it is known as _____.

- a) Function recursion
- b) Tail recursion**
- c) Last recursion
- d) Local Recursion

257. parameter passing (by value or by reference) is similar to PASCAL.

- a) C++**
- b) COBOL
- c) JAVA
- d) FORTRAN

258. Given a stack of n items, how many POP and PUSH operations need to be performed to remove the item at its bottom?

- a) 0 POP operation and 0 PUSH operation
- b) 1 POP operation and 1 PUSH operation
- c) n POP operations and n PUSH operations
- d) n POP operations and $n-1$ PUSH operations**

Best of Luck

For Free help, more Mid/Final term Files and for Exam Preparation You Can Join:

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